

Optimal majorization uncertainty relations for two-outcome qubit measurements

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Entropic uncertainty relations such as the Maassen–Uffink bound compress the incompatibility of two measurements into a single scalar inequality. Majorization offers a finer description: a single state-independent vector ω that majorizes the outcome statistics of every state yields, at once, lower bounds on every Schur-concave uncertainty measure. We determine this vector in closed form for pairs of two-outcome qubit measurements. For projective measurements with maximal overlap $c = \max_{i,j} |\langle m_i | n_j \rangle|^2$ we prove that the optimal direct-sum bound is $\omega_{\oplus} = (1, \sqrt{c}, 1 - \sqrt{c}, 0)$ and the optimal tensor-product bound is $\omega_{\otimes} = (w, 1 - w, 0, 0)$ with $w = (1 + \sqrt{c})^2/4$. Both bounds are the least upper bounds of the set of achievable statistics in the majorization lattice, yet neither is attained by any state unless the measurements commute; we identify the distinct families of states that saturate each partial sum. The resulting Shannon bound $h(\sqrt{c})$ is asymptotically tight for nearly compatible measurements and outperforms Maassen–Uffink whenever the Bloch angle between the measurement axes is below 81.3° . We then extend the direct-sum result to arbitrary (unsharp and biased) two-outcome qubit POVMs, and, via Jordan’s lemma, to two-outcome projective measurements in any finite dimension.

I. INTRODUCTION

The uncertainty principle [1, 2] expresses the impossibility of preparing a quantum state that yields sharp outcomes for two incompatible measurements. Its information-theoretic formulation replaces variances by entropies of the outcome distributions p and q of two measurements M and N [3–5]. The best known example is the Maassen–Uffink (MU) relation [5]

$$H(p) + H(q) \geq -\log_2 c, \quad c = \max_{i,j} |\langle m_i | n_j \rangle|^2, \quad (1)$$

which has become a workhorse of quantum cryptography and quantum information theory [6, 7].

A scalar inequality such as (1) is clean and effective but necessarily discards information: it constrains one particular functional of the pair (p, q) , and a different uncertainty measure requires a different relation. Majorization [8] provides a structural alternative. One seeks a single state-independent probability vector ω such that

$$p(\rho) \oplus q(\rho) \prec \omega \quad \text{or} \quad p(\rho) \otimes q(\rho) \prec \omega \quad \forall \rho, \quad (2)$$

so that every Schur-concave function Φ —Shannon, Rényi or Tsallis entropies, among many others—inherits the bound $\Phi(p \oplus q) \geq \Phi(\omega)$ automatically. This programme was initiated by Partovi [9] and developed into *universal uncertainty relations* by Friedland, Gheorghiu and Gour [10] (tensor-product form) and by Puchała, Rudnicki and Życzkowski [11, 12] (tensor-product and direct-sum forms). These works provide general constructions of valid bounds, but explicit, provably optimal bounds together with a characterization of which states come closest to them are available only in special cases.

In this paper we give a complete answer for the simplest non-trivial setting: a pair of two-outcome measurements on a qubit. Our results are:

- (i) A geometric description of the achievable set $\wp(M, N) = \{p(\rho) \oplus q(\rho)\}$ as an affine image of an ellipse whose boundary is traced by pure states (Lemma 2).
- (ii) Closed-form optimal direct-sum and tensor-product bounds $\omega_{\oplus}$ and $\omega_{\otimes}$ for projective measurements (Theorems 1 and 2), with a proof that neither is attained unless $[M, N] = 0$ and an identification of the saturating states (Proposition 1).
- (iii) Entropic consequences: the direct-sum Shannon bound $h(\sqrt{c})$ always dominates the tensor-product one, beats MU for Bloch angles below 81.3° , and is asymptotically tight as $c \rightarrow 1$ (Sec. V).
- (iv) The optimal direct-sum bound for arbitrary two-outcome qubit POVMs (Theorem 3) and for two-outcome projective measurements in any dimension (Theorem 4).

All statements are accompanied by numerical checks; the code is available with this manuscript (Appendix A).

II. PRELIMINARIES

A. Majorization

For $x \in \mathbb{R}^d$ let $x^\downarrow$ denote its entries in non-increasing order and $L_k(x) = \sum_{i=1}^k x_i^\downarrow$ its Lorenz partial sums. For x, y with equal total, x is *majorized* by y , written $x \prec y$, if $L_k(x) \leq L_k(y)$ for all k [8]. A function Φ is Schur-concave if $x \prec y$ implies $\Phi(x) \geq \Phi(y)$. Every function

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of the form $\Phi(x) = \sum_i \phi(x_i)$ with ϕ concave is Schur-concave (Hardy–Littlewood–Pólya–Karamata); so are all Rényi entropies H_α , $\alpha \in [0, \infty]$.

Restricted to non-increasing vectors of fixed total, $\prec$ is a partial order and, in fact, a lattice [13].

Definition 1. Let A be a set of vectors with common total. A vector ω is an *optimal majorization bound* for A if $a \prec \omega$ for all $a \in A$, and $\omega \prec \omega'$ for every other ω' with this property; i.e., $\omega = \sup A$ in the majorization lattice.

Lemma 1 ([13]). Let $S_k = \sup_{a \in A} L_k(a)$. Then $\sup A$ is the vector whose Lorenz curve is the least concave majorant of $k \mapsto S_k$ (with $S_0 = 0$). In particular, if $\omega_k = S_k - S_{k-1}$ is already non-increasing, then $\omega = \sup A$.

In all cases treated below the second, simpler statement applies. The quantities S_k have a transparent meaning: S_k is the largest total probability that any state can concentrate on k outcomes.

B. Two-outcome qubit measurements

A qubit state is $\rho = \frac{1}{2}(\mathbb{I} + \vec{r} \cdot \vec{\sigma})$ with $|\vec{r}| \leq 1$. A projective two-outcome measurement is $M = \{P_\pm\}$ with $P_\pm = \frac{1}{2}(\mathbb{I} \pm \hat{m} \cdot \vec{\sigma})$, $|\hat{m}| = 1$, and similarly $N = \{Q_\pm\}$ with axis $\hat{n}$. The outcome distributions are

$$p = \left(\frac{1+x}{2}, \frac{1-x}{2}\right), \quad q = \left(\frac{1+y}{2}, \frac{1-y}{2}\right), \quad x = \hat{m} \cdot \vec{r}, \quad y = \hat{n} \cdot \vec{r}. \quad (3)$$

The overlaps are $|\langle m_\pm | n_\pm \rangle|^2 = \frac{1}{2}(1 + \hat{m} \cdot \hat{n})$ and $|\langle m_\pm | n_\mp \rangle|^2 = \frac{1}{2}(1 - \hat{m} \cdot \hat{n})$, hence

$$c = \frac{1}{2}(1 + |\hat{m} \cdot \hat{n}|) = \cos^2 \frac{\theta}{2}, \quad \cos \theta = |\hat{m} \cdot \hat{n}|, \quad \theta \in [0, \frac{\pi}{2}]. \quad (4)$$

Relabelling the outcomes of N maps $\hat{n} \mapsto -\hat{n}$ and does not affect any majorization statement, so we may and shall assume $\hat{m} \cdot \hat{n} = \cos \theta \geq 0$. Note that $\frac{1}{2} \leq c \leq 1$ and $\sqrt{c} \geq 1/\sqrt{2} > 1 - \sqrt{c}$.

A general two-outcome POVM $\{E_0, E_1 = \mathbb{I} - E_0\}$ can always be written as

$$E_0 = \frac{1}{2}[(1 + \mu)\mathbb{I} + \eta \hat{a} \cdot \vec{\sigma}], \quad \eta \geq 0, \quad |\mu| + \eta \leq 1, \quad (5)$$

the constraint being equivalent to $0 \leq E_0 \leq \mathbb{I}$. Here η is the sharpness and μ the bias; projective measurements correspond to $\eta = 1$, $\mu = 0$.

III. GEOMETRY OF THE ACHIEVABLE SET

The statistics (3) depend on $\vec{r}$ only through (x, y) .

Lemma 2. For non-parallel axes, the set $\{(\hat{m} \cdot \vec{r}, \hat{n} \cdot \vec{r}) : |\vec{r}| \leq 1\}$ is the filled ellipse

$$\mathcal{E}_\theta = \{(x, y) : x^2 - 2xy \cos \theta + y^2 \leq \sin^2 \theta\}. \quad (6)$$

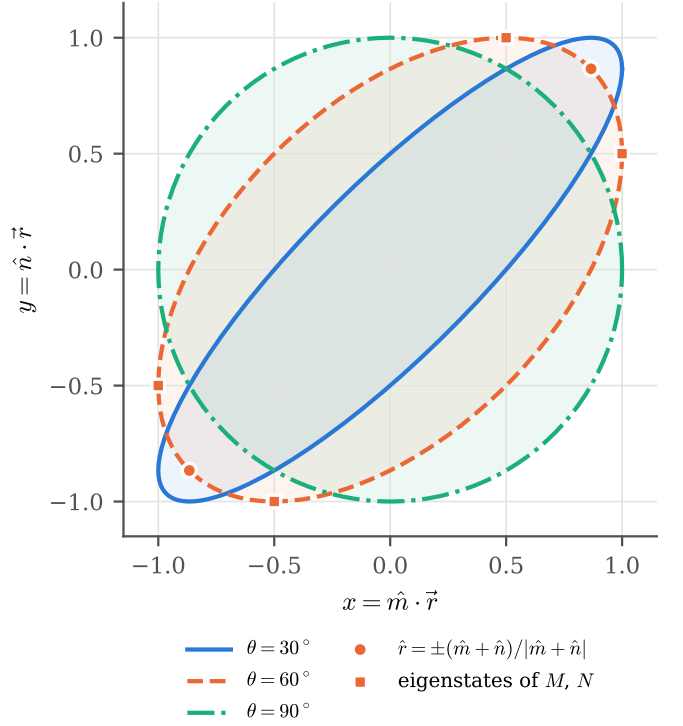


FIG. 1. The ellipse $\mathcal{E}_\theta$ of achievable expectation values $(x, y) = (\langle \hat{m} \cdot \vec{\sigma} \rangle, \langle \hat{n} \cdot \vec{\sigma} \rangle)$ for three Bloch angles. Markers (for $\theta = 60^\circ$) show the two families of states that saturate the partial sums of $\omega_\oplus$: eigenstates of M or N (squares; $k = 1, 3$) and bisector states $\hat{r} \propto \pm(\hat{m} + \hat{n})$ (circles; $k = 2$). No single point saturates all partial sums (Proposition 1).

Its boundary consists exactly of the images of the pure states whose Bloch vectors lie in $\text{span}\{\hat{m}, \hat{n}\}$. Moreover, for all $\alpha, \beta \in \mathbb{R}$,

$$\max_{|\vec{r}| \leq 1} (\alpha x + \beta y) = |\alpha \hat{m} + \beta \hat{n}|. \quad (7)$$

Proof. Write $\vec{r} = \text{VisionRes}_\parallel + \text{VisionRes}_\perp$ with $\text{VisionRes}_\parallel \in \text{span}\{\hat{m}, \hat{n}\}$. Only $\text{VisionRes}_\parallel$ contributes to (x, y) , and $(x, y) = G\xi$ where ξ are the coefficients of $\text{VisionRes}_\parallel$ in the basis $\{\hat{m}, \hat{n}\}$ and $G = \begin{pmatrix} 1 & \cos \theta \\ \cos \theta & 1 \end{pmatrix}$ is the Gram matrix. Then $|\text{VisionRes}_\parallel|^2 = \xi^T G \xi = (x, y) G^{-1} (x, y)^T \leq 1$, which is the stated inequality since $\det G = \sin^2 \theta$. Equality holds iff $\text{VisionRes}_\perp = 0$ and $|\vec{r}| = 1$. Eq. (7) is Cauchy–Schwarz: $\alpha x + \beta y = \vec{r} \cdot (\alpha \hat{m} + \beta \hat{n})$. $\square$

Consequently the achievable set $\wp(M, N) = \{p(\rho) \oplus q(\rho)\} \subset \mathbb{R}^4$ is the image of $\mathcal{E}_\theta$ under the affine map $(x, y) \mapsto \frac{1}{2}(1 + x, 1 - x, 1 + y, 1 - y)$: it is convex and compact, and its extreme points are the statistics of pure states in the plane of the two axes (Fig. 1). Any uncertainty measure that is concave in (x, y) , such as $H(p) + H(q)$, is therefore minimized on $\partial \mathcal{E}_\theta$, and the optimization problems below are effectively one-dimensional.

IV. PROJECTIVE MEASUREMENTS

A. Direct-sum bound

Theorem 1. *Let M, N be projective qubit measurements with maximal overlap c . For every state ρ ,*

$$p(\rho) \oplus q(\rho) \prec \omega_{\oplus} := (1, \sqrt{c}, 1 - \sqrt{c}, 0), \quad (8)$$

and $\omega_{\oplus}$ is the optimal majorization bound for $\wp(M, N)$.

Proof. We compute $S_k = \max_{\rho} L_k(p \oplus q)$ for $k = 1, \dots, 4$. Clearly $S_1 \leq 1$ and $S_4 = 2$. Since all entries are non-negative, $S_3 = 2 - \min_{\rho} \min_i (p \oplus q)_i \leq 2$. Both bounds are attained by an eigenstate of M , for which $p = (1, 0)$. For $k = 2$, the two largest entries either belong to the same distribution, with sum 1, or are of the form $p_i + q_j$. By Lemma 2,

$$p_i + q_j = 1 + \frac{1}{2}(\pm x \pm y) \leq 1 + \frac{1}{2} \max\{|\hat{m} + \hat{n}|, |\hat{m} - \hat{n}|\}. \quad (9)$$

With $\hat{m} \cdot \hat{n} = \cos \theta \geq 0$ the maximum is $|\hat{m} + \hat{n}| = \sqrt{2(1 + \cos \theta)} = 2 \cos \frac{\theta}{2} = 2\sqrt{c}$, so

$$S_2 = 1 + \sqrt{c}, \quad (10)$$

attained by the bisector state $\vec{r} = (\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$. This is the qubit case of the Landau–Pollak-type inequality $p_i + q_j \leq 1 + \sqrt{c}$ underlying Deutsch’s relation [3]. The increments of $(S_1, S_2, S_3, S_4) = (1, 1 + \sqrt{c}, 2, 2)$ give $\omega_{\oplus}$, which is non-increasing because $1 \geq \sqrt{c} \geq 1 - \sqrt{c} \geq 0$. Lemma 1 completes the proof. $\square$

Proposition 1. *There is a state with $(p \oplus q)^{\downarrow} = \omega_{\oplus}$ if and only if $c = 1$, i.e., M and N commute. For $c < 1$, the maxima S_1, S_3 are attained only by the eigenstates of M and N , and S_2 only by the bisector states $\vec{r} = \pm(\hat{m} + \hat{n})/|\hat{m} + \hat{n}|$.*

Proof. $(p \oplus q)_1^{\downarrow} = 1$ forces ρ to be an eigenstate of M or N , say $\vec{r} = \pm \hat{m}$. Then $q = (\frac{1 \pm \cos \theta}{2}, \frac{1 \mp \cos \theta}{2})$ and $(p \oplus q)^{\downarrow} = (1, c, 1 - c, 0)$, which equals $\omega_{\oplus}$ iff $c = \sqrt{c}$, i.e., $c = 1$. The uniqueness of the maximizer of S_2 follows from the equality condition in Cauchy–Schwarz and the strict inequality $|\hat{m} - \hat{n}| < |\hat{m} + \hat{n}|$ for $\theta < \pi/2$ (for $\theta = \pi/2$ both bisectors $\pm \hat{m} \pm \hat{n}$ qualify). $\square$

Proposition 1 makes precise the sense in which $\omega_{\oplus}$ is a *description* of the achievable set rather than the statistics of a particular state: it is the supremum of $\wp(M, N)$ in the majorization lattice, and its partial sums are saturated by two disjoint families of states (Fig. 2). The gap between $(1, c, 1 - c, 0)$ and $\omega_{\oplus}$ quantifies the price one pays for demanding certainty about M .

B. Tensor-product bound

Theorem 2. *For every state ρ ,*

$$p(\rho) \otimes q(\rho) \prec \omega_{\otimes} := (w, 1 - w, 0, 0), \quad (11)$$

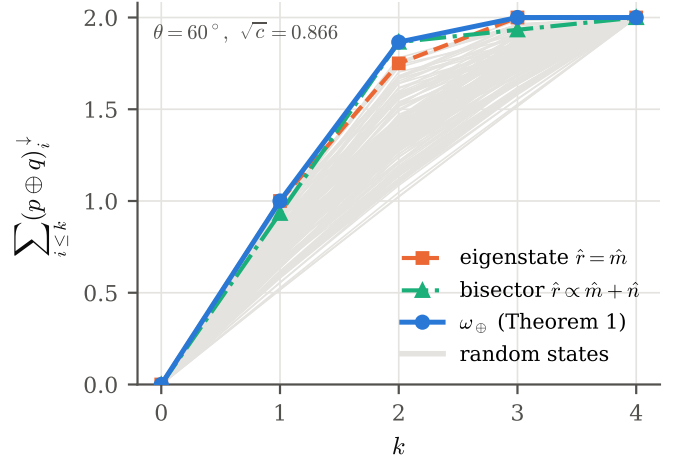


FIG. 2. Lorenz curves of $p \oplus q$ for $\theta = 60^\circ$. Grey: 150 random states. The optimal bound $\omega_{\oplus}$ (blue) is the upper envelope; it touches the eigenstate curve at $k = 1, 3$ and the bisector-state curve at $k = 2$, but no single state reaches it.

where $w = \frac{1}{4}(1 + \sqrt{c})^2$, and $\omega_{\otimes}$ is the optimal majorization bound for $\{p(\rho) \otimes q(\rho)\}$.

Proof. By the arithmetic–geometric mean inequality and Eq. (10), $p_i q_j \leq (\frac{p_i + q_j}{2})^2 \leq (\frac{1 + \sqrt{c}}{2})^2 = w$, with equality for the bisector state, for which $p_+ = q_+ = (1 + \sqrt{c})/2$. Hence $S_1 = w$. For an eigenstate of M , $p \otimes q = (q_+, q_-, 0, 0)$, so $S_2 = S_3 = S_4 = 1$. Since $\sqrt{c} \geq 1/\sqrt{2}$ we have $w \geq \frac{1}{2}$, so the increments $(w, 1 - w, 0, 0)$ are non-increasing, and Lemma 1 applies. $\square$

As before, $\omega_{\otimes}$ is attained only if $c = 1$: at the bisector state $L_2(p \otimes q) = p_+ = (1 + \sqrt{c})/2 < 1$ whenever $c < 1$.

V. ENTROPIC CONSEQUENCES

Let $h(t) = -t \log_2 t - (1 - t) \log_2 (1 - t)$ be the binary entropy.

Corollary 1. *For every concave ϕ on $[0, 1]$, $\sum_i \phi(p_i) + \sum_j \phi(q_j) \geq \phi(1) + \phi(\sqrt{c}) + \phi(1 - \sqrt{c}) + \phi(0)$. In particular,*

$$H(p) + H(q) \geq h(\sqrt{c}). \quad (12)$$

Corollary 2. *For every $\alpha \in [0, \infty]$,*

$$H_{\alpha}(p) + H_{\alpha}(q) \geq H_{\alpha}(\omega_{\otimes}) = \frac{\log_2[w^{\alpha} + (1 - w)^{\alpha}]}{1 - \alpha}. \quad (13)$$

For $\alpha = 1$ this reads $H(p) + H(q) \geq h(w)$; for $\alpha = \infty$ it gives $H_{\infty}(p) + H_{\infty}(q) \geq -2 \log_2 \frac{1 + \sqrt{c}}{2}$, which is tight (bisector state); since $H \geq H_{\infty}$, it strengthens Deutsch’s Shannon bound [3], which has the same right-hand side.

The tensor form is needed for Rényi entropies because $H_{\alpha}(p \otimes q) = H_{\alpha}(p) + H_{\alpha}(q)$ for all α , whereas only $\alpha = 1$ is additive over direct sums. For the Shannon entropy, however, the direct sum is always better:

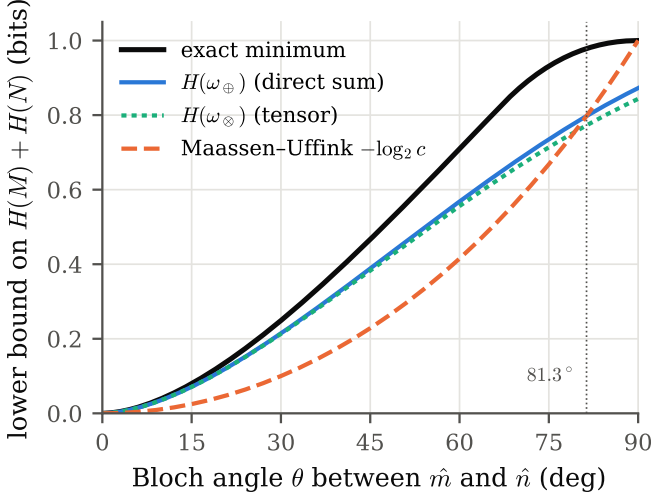


FIG. 3. Lower bounds on $H(M) + H(N)$ as a function of the Bloch angle θ ($c = \cos^2 \frac{\theta}{2}$). Black: exact minimum. Blue: direct-sum bound $h(\sqrt{c})$. Green (dotted): tensor-product bound $h(w)$. Orange (dashed): Maassen-Uffink. The direct-sum bound is stronger than MU for $\theta < 81.3^\circ$ (dotted vertical line).

Proposition 2. $h(\sqrt{c}) \geq h(w)$, with equality iff $c = 1$.

Proof. Both $\sqrt{c}$ and w lie in $[\frac{1}{2}, 1]$ where h is decreasing, and $w - \sqrt{c} = \frac{1}{4}(1 - \sqrt{c})^2 \geq 0$. $\square$

Figure 3 compares the bounds with the exact minimum of $H(p) + H(q)$, obtained by a one-dimensional optimization over the boundary of $\mathcal{E}_\theta$ [14, 15]. For mutually unbiased measurements ($\theta = 90^\circ$, $c = \frac{1}{2}$) MU is tight (1 bit), while $h(1/\sqrt{2}) \approx 0.872$ and $h(w) \approx 0.844$. As the measurements become more compatible, however, the majorization bounds take over: $h(\sqrt{c}) > -\log_2 c$ for $\sqrt{c} > t^* \approx 0.7585$, i.e., for Bloch angles $\theta < 81.3^\circ$ (the tensor bound overtakes MU at $\theta < 79.3^\circ$).

The advantage becomes unbounded as $c \rightarrow 1$. Writing $\varepsilon = 1 - \sqrt{c} \approx \theta^2/8$,

$$-\log_2 c \approx \frac{2\varepsilon}{\ln 2}, \quad h(\sqrt{c}) \approx \varepsilon \log_2 \frac{e}{\varepsilon}, \quad (14)$$

so the ratio $h(\sqrt{c})/(-\log_2 c)$ diverges logarithmically. More strongly, the direct-sum bound is asymptotically *tight*. To leading order the exact minimum is attained near the pure state whose Bloch vector bisects $\hat{m}$ and $\hat{n}$, $\hat{r} \cdot \hat{m} = \hat{r} \cdot \hat{n} = \cos \frac{\theta}{2}$; there $1 - p_+ = 1 - q_+ = \varepsilon/2$, which yields $H(p) + H(q) \approx \varepsilon \log_2 \frac{2e}{\varepsilon}$, whose ratio to $h(\sqrt{c})$ tends to 1. Numerically, the ratio of the exact minimum to $h(\sqrt{c})$ is 1.09 at $\theta = 0.1$ and 1.047 at $\theta = 0.003$, while the corresponding ratio for MU grows from 4.2 to 7.7. Note that the eigenstates of M , although they saturate S_1 and S_3 , are *not* the entropy minimizers in this regime: they give $h(c)$, which is asymptotically twice the minimum.

VI. GENERAL TWO-OUTCOME QUBIT POVMS

Let $M = \{E_0, E_1\}$ with parameters $(\mu, \eta, \hat{a})$ and $N = \{F_0, F_1\}$ with $(\nu, \zeta, \hat{b})$, as in Eq. (5). Then $p = \frac{1}{2}(1 + \mu + \eta \hat{a} \cdot \vec{r}, 1 - \mu - \eta \hat{a} \cdot \vec{r})$, and similarly for q .

Theorem 3. *Define*

$$\kappa = \max\{|\mu| + \eta, |\nu| + \zeta\}, \quad (15)$$

$$\Lambda = \max_{s,t=\pm 1} [s\mu + t\nu + |s\eta \hat{a} + t\zeta \hat{b}|]. \quad (16)$$

Then $p(\rho) \oplus q(\rho) \prec \omega_\oplus$ for all ρ , where

$$\omega_\oplus = \frac{1}{2}(1 + \kappa, 1 + \Lambda - \kappa, 1 + \kappa - \Lambda, 1 - \kappa) \quad (17)$$

is the optimal majorization bound.

Proof. The largest single outcome probability is $\max_\rho \frac{1}{2}(1 + |\mu + \eta \hat{a} \cdot \vec{r}|) = \frac{1}{2}(1 + |\mu| + \eta)$ for M , so $S_1 = \frac{1}{2}(1 + \kappa)$; likewise the smallest is $\frac{1}{2}(1 - \kappa)$, so $S_3 = 2 - \frac{1}{2}(1 - \kappa)$. For S_2 , a pair within one distribution sums to 1, while a cross pair gives, by Eq. (7),

$$\begin{aligned} \max_\rho [1 + \frac{1}{2}(s(\mu + \eta \hat{a} \cdot \vec{r}) + t(\nu + \zeta \hat{b} \cdot \vec{r}))] \\ = 1 + \frac{1}{2}[s\mu + t\nu + |s\eta \hat{a} + t\zeta \hat{b}|]. \end{aligned} \quad (18)$$

Choosing $s = \text{sgn } \mu$, $t = \text{sgn } \nu$ shows $\Lambda \geq 0$, so $S_2 = 1 + \frac{1}{2}\Lambda$. The increments of (S_k) are the entries of Eq. (17). They are non-increasing iff $\kappa \leq \Lambda \leq 2\kappa$. The upper inequality follows from the triangle inequality, $\Lambda \leq |\mu| + |\nu| + \eta + \zeta \leq 2\kappa$. For the lower one, let ρ^* attain S_1 via an outcome of, say, M ; then $S_2 \geq S_1 + \max_j q_j(\rho^*) \geq S_1 + \frac{1}{2}$, i.e., $\Lambda \geq \kappa$. Lemma 1 completes the proof. $\square$

For projective measurements $\kappa = 1$, $\Lambda = 2\sqrt{c}$ and Eq. (17) reduces to Eq. (8). For unbiased measurements of equal sharpness ($\mu = \nu = 0$, $\eta = \zeta$) one finds $\omega_\oplus = \eta(1, \sqrt{c}, 1 - \sqrt{c}, 0) + \frac{1-\eta}{2}(1, 1, 1, 1)$, the noisy version of the sharp bound, as expected from the fact that such measurements are sharp measurements followed by a depolarizing channel. For unequal sharpness or nonzero bias, $\omega_\oplus$ is not of this form, and the relevant geometric quantity is the length $|\eta \hat{a} \pm \zeta \hat{b}|$ rather than the overlap c . As a corollary, $H(p) + H(q) \geq -\sum_k \omega_k \log_2 \omega_k$ for all two-outcome qubit POVMs. Figure 4 shows that this bound tracks the exact minimum over the whole range of sharpness, becoming exact in the trivial limit $\eta \rightarrow 0$.

VII. BEYOND QUBITS: TWO-OUTCOME PROJECTIVE MEASUREMENTS IN DIMENSION d

The qubit analysis already contains the general two-outcome projective case.

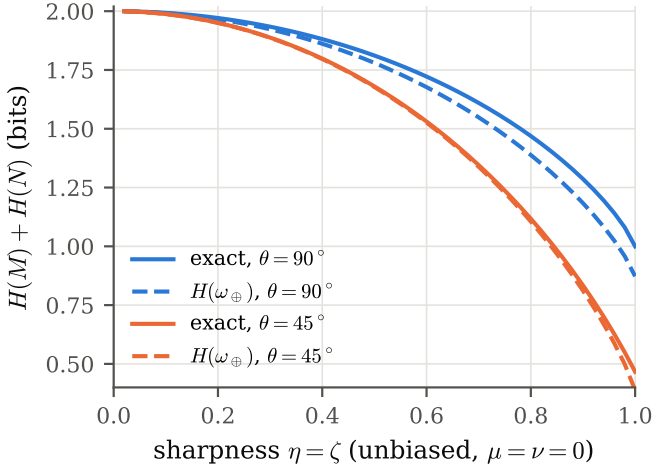


FIG. 4. Unsharp measurements ($\mu = \nu = 0$, $\eta = \zeta$): exact minimum of $H(M) + H(N)$ (solid) and the bound from Theorem 3 (dashed) for Bloch angles 90° and 45° .

Theorem 4. Let $M = \{P, \mathbb{I} - P\}$ and $N = \{Q, \mathbb{I} - Q\}$ be projective measurements on $\mathbb{C}^d$ and $c = \max_{i,j} \|P_i Q_j\|_\infty^2$. Then Theorems 1 and 2 hold verbatim.

Proof sketch. By Jordan’s lemma [16], $\mathbb{C}^d$ decomposes into mutually orthogonal subspaces of dimension one or two, each invariant under P and Q . On a two-dimensional block b , P and Q act as rank-one projectors, i.e., as a pair of qubit measurements with overlap c_b ; on a one-dimensional block they have a common eigenvector ($c_b = 1$). Because P and Q are block diagonal, $p(\rho) \oplus q(\rho) = \sum_b \lambda_b p_b \oplus q_b$ is a convex combination of qubit statistics. Since $\omega_\oplus(c)$ increases in the majorization order with c and $\{v : v \prec \omega\}$ is convex, every $p \oplus q$ is majorized by $\omega_\oplus(\max_b c_b)$, and $\max_b c_b = c$. Optimality follows by embedding the maximizing qubit states of the block with $c_b = c$. For the tensor product we use again $p_i q_j \leq (\frac{p_i + q_j}{2})^2$ with $p_i + q_j \leq 1 + \sqrt{c}$. $\square$

More generally, for arbitrary POVMs the direct-sum partial sums are $S_k = \max_{|J|=k} \lambda_{\max}(\sum_{j \in J} F_j)$, where the F_j run over all effects of both measurements. This gives an exact numerical procedure that we use to verify Theorems 3 and 4 independently (Appendix A).

VIII. DISCUSSION AND OUTLOOK

We have shown that, for two-outcome qubit measurements, the achievable set of outcome statistics admits a compact majorization description: a single four-component vector, determined by the overlap c (or, for POVMs, by the parameters κ and Λ), that is the least upper bound of all achievable statistics. The bound captures every Schur-concave uncertainty measure simultaneously, is never attained by a single state when the

measurements are incompatible, and yields a Shannon relation that is asymptotically optimal in the nearly-compatible regime, precisely where MU is weakest.

Several questions remain open. (i) *Tensor bounds for POVMs.* The AM–GM step of Theorem 2 is no longer tight for biased or unequally sharp measurements; the optimal $\omega_\otimes$ requires maximizing a quadratic form over $\mathcal{E}_\theta$. (ii) *Naimark dilation.* Every two-outcome POVM is a projective measurement on a qubit coupled to an ancilla [17]. Combined with Theorem 4, this suggests relating (κ, Λ) to the principal angles of the dilated projectors, restricted to product inputs. (iii) *More measurements and outcomes.* For three or more measurements, or more outcomes, the least concave majorant in Lemma 1 is no longer automatically trivial, and the lattice structure of the bound becomes genuinely relevant. (iv) *Beyond quantum mechanics.* The approach replaces scalar summaries by comparisons of full distributions. The same philosophy may be useful for other ordering problems in information processing, for instance in comparing the concentration of attention distributions in machine-learning models.

ACKNOWLEDGMENTS

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Appendix A: Numerical methods and code

All results were checked with the accompanying Python code (code/):

- **majorization.py:** POVM parametrization, Lorenz curves, the lattice supremum via least concave majorants [13], the analytic bounds of Theorems 1–4, and the exact eigenvalue characterization $S_k = \max_{|J|=k} \lambda_{\max}(\sum_{j \in J} F_j)$.
- **verify_bounds.py:** compares the closed forms with (a) lattice suprema over 1.8×10^5 grid states (maximal deviation $< 10^{-4}$), and (b) the eigenvalue characterization for 200 random POVM pairs and 200 random projector pairs in dimensions 2–6 (deviation $< 10^{-14}$); it also checks Proposition 1 and Corollaries 1–2.
- **make_figures.py:** generates Figs. 1–4.

The exact entropy minima are obtained by a dense grid over the great circle in $\text{span}\{\hat{m}, \hat{n}\}$ refined by Nelder–Mead. This reduction is justified by Lemma 2: the Shannon entropy is concave in (x, y) and is therefore minimized on the boundary of $\mathcal{E}_\theta$. For unbiased measurements, moving $\vec{r}$ outwards makes both p and q more ordered, so the same holds for every Schur-concave functional, including H_∞ .

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